Botanical Journal of the Linnean Society, 2018, **XX**, 1–19. With 6 figures.

Morphological and harmomegathic characters of pollen of *Gnetum* species (Gnetaceae) in Thailand: studies with light and scanning electron microscopy

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Received 2 August 2017; revised 11 December 2017; accepted for publication 27 April 2018

The morphological plasticity that can occur in pollen of *Gnetum* due to harmomegathy demands a comparative overview of pollen characters in *Gnetum* when studying systematics, particularly when fossil pollen grains are taken into account. Using LM and SEM, we studied acetolyzed and dry and rehydrated non-acetolyzed pollen grains of eight *Gnetum* spp. in Thailand. They are subspheroidal, inaperturate, microechinate monads with blunt-ended microechini. In acetolyzed and non-acetolyzed pollen grains, the raised exine regions and the ridge–like folds appear to be characteristic of the dehydrated state, whereas a psilate surface between microechini is representative of the hydrated state. Although we were able to measure and record some variation in average values of certain features of all the pollen studied and in some characters of the dehydrated specimens, the general morphological similarity of all the examined *Gnetum* pollen makes it difficult to establish any systematic significance. Nonetheless, their continuous tectum supports the molecular phylogenetic information that suggests Asian *Gnetum* spp. are monophyletic. This study presents the first report of the pollen morphology of *Gnetum microcarpum* Blume and *G. tenuifolium* Ridl. © 2018 The Linnean Society of London, *Biological Journal of the Linnean Society*, 2018, **XX**, 000–000.

ADDITIONAL KEYWORDS: exine – *Gnetum* – harmomegathy – palynology.

INTRODUCTION

Gnetales are a monophyletic order, comprising three families of gymnosperms, Ephedraceae, Gnetaceae and Welwitschiaceae. *Gnetum* L. belongs to Gnetaceae and comprises evergreen dioecious (or rarely monoecious) trees, shrubs and woody climbers, growing in moist tropical and subtropical regions of Asia, Africa and South America (Markgraf, 1929; Crane, 1985; Kubitzki, 1990; Price, 1996; Friis, Crane & Pedersen, 2011). There are *c*. 43 species: four in Africa, seven in South America and *c*. 32 in Asia (Biye, Balkwill & Cron, 2014; Hou *et al.*, 2015).

The systematic relationships among *Gnetum* spp. have been well studied (Griffith, 1859; Markgraf, 1929; Carlquist, 1996; Won & Renner, 2005, 2006; Hou *et al.*, 2015). Based on morphological data, two sections have been established, including *Gnetum* (synonyms: sections *Erecta* Griff. and *Gnemonomorphi* Markgr.) and *Scandentia* Griff. (synonym: section *Cylindrostachys* Markgr.). *Gnetum* section *Gnetum* includes three subsections, *Araeognemones* Markgr. (American taxa), *Micrognemones* Markgr. (African taxa) and *Eugnemones* Markgr. (arborescent Asian taxa). *Gnetum* section *Scandentia* (lianoid Asian taxa) is further divided into two subsections, *Stipitati* Markgr. and *Sessiles* Markgr. (Griffith, 1859; Markgraf, 1929; Price, 1996). Markgraf (1929) proposed that arborescent *Gnetum* taxa are ancestral and had diverged earlier than other species in the genus. The

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molecular analyses of Hou *et al.* (2015) suggested that section *Gnetum* is paraphyletic and section *Scandentia* is monophyletic. However, the subsections of section *Gnetum* were monophyletic, whereas those of section *Scandentia* were not. Their results also provided strong support for the existence of three geographically defined major clades of *Gnetum*: the American, African and Asian clades.

The pollen morphology of *Gnetum* has been studied for more than 80 years (Wodehouse, 1935; Erdtman, 1965; Gullvåg, 1966; Hesse, 1980; Orel, Kuprianova & Golubeva, 1986; Gillespie & Nowicke, 1994; Osborn, 2000; Yao *et al.*, 2004; Ickert-Bond & Renner, 2016; Tekleva, 2016). The pollen has been generally described as small, spheroidal or subspheroidal, inaperturate and microechinate (Erdtman, 1965; Gillespie & Nowicke, 1994; Osborn, 2000; Yao *et al.*, 2004; Tekleva, 2016). However, a pore-like region was mentioned in one previous study (Erdtman, 1965). The pollen wall ultrastructure showed a thin tectum, granular infratectum, lamellate endexine and an indistinct or absent foot layer (Osborn, 2000; Yao *et al.*, 2004; Tekleva, 2016). Based on differences observed in the tectal ornamentations by light microscopy (LM) alone, Erdtman (1954, 1965) divided *Gnetum* pollen into three types: spinulose (Asian taxa), pilate (Neotropical species) and insulous (African species). Different results were obtained from scanning electron microscopy (SEM) analyses as only two pollen types were recognized. These are the Asian type, represented by a continuous tectum with microechini, and the American and African type, characterized by a discontinuous tectum forming plate-like regions or islands with more numerous microechini (Gillespie & Nowicke, 1994; Tekleva, 2016).

The raised exine region and ridge (or fold) around the echinus or echini, forming a network-like pattern, are interesting characters of pollen from Asian *Gnetum* (Gillespie & Nowicke, 1994; Yao *et al.*, 2004; Tekleva, 2016). Gillespie & Nowicke (1994) observed the differences in the tectum surfaces of pollen samples from *G. latifolium* Blume and stated that the variation of folding found was influenced by sample preparation procedures. However, in a natural environment, pollen morphology can change to accommodate the pressure changes within pollen grains in response to wet or dry conditions (Wodehouse, 1935; Hesse *et al.*, 2009) and numerous studies have demonstrated the plasticity of pollen morphology due to harmomegathy (Payne, 1972; Pacini, 1990; Scotland, Barnes & Blackmore, 1990; Gillespie & Nowicke, 1994; Hesse, 2000; Hesse *et al.*, 2001; Halbritter & Hesse, 2004; Hesse *et al.*, 2009; Halbritter, Weber & Hesse 2010; Katifori *et al.*, 2010; Volkova, Severova & Polevova, 2013). This causes remarkable differences in the appearance of pollen even from the same plant species, leading to problems when making comparative interpretations, particularly in the fossil record.

In Thailand, eight *Gnetum* taxa have been observed: *G. cuspidatum* Blume; *G. latifolium* var. *funiculare* (Blume) Markgr.; *G. leptostachyum* Blume; *G. gnemon* L. var. *tenerum* Markgr.; *G. macrostachyum* Hook.f.; *G. microcarpum* Blume; *G. montanum* Markgr. and *G. tenuifolium* Ridl. (Phengklai, 1973). Besides the pollen morphology of *G. microcarpum* and *G. tenuifolium*, other pollen morphologies have been reported (Gillespie & Nowicke, 1994; Osborn, 2000; Yao *et al.*, 2004; Tekleva, 2016). However, these morphological characterizations were based on examinations of acetolyzed or dry pollen grains from herbaria. No hydrated or rehydrated pollen grains treated with different methods have been previously examined. Since *Gnetum* pollen exhibits morphological flexibility depending on the environmental conditions or even the preparation technique used, palynological data, particularly from fossil *Gnetum* pollen, can be misinterpreted. Therefore, this present study aimed to document the pollen morphology of *Gnetum* pollen in Thailand and harmomegathic characters in order to provide a proper perspective on the comparative morphology of these pollen to establish their systematic significance.

MATERIAL AND METHODS

Pollen from all the *Gnetum* spp. recorded in Thailand (Phengklai, 1973) were studied. Except for pollen from two specimens of *G. microcarpum* and one specimen of *G. macrostachyum*, which were collected from Kho Hong Hill, Songkhla province, southern Thailand, the pollen samples of all species were taken from the Herbarium, Biology Department, Faculty of Science, Prince of Songkla University (PSU), the Herbarium of the Department of Biology, Faculty of Science, Chang-Mai University (CMU) and the Queen Sirikit Botanic Garden Herbarium (QBG), Botanical Garden Organization, Thailand (see Appendix). To ensure the correct identification of the pollen from *G. microcarpum*, collected plant samples were compared with specimens deposited in the PSU herbarium. For LM study, pollen grains were acetolyzed and mixed with glycerin before mounting on slides (Erdtman, 1969). The pollen size, outline, exine thickness and apertural condition were investigated. The pollen was then photographed under light microscopy, BX43, Olympus. For SEM analysis, two types of pollen samples were examined: acetolyzed and non-acetolyzed. All the non-acetolyzed pollen samples were studied in dry and rehydrated conditions. The acetolyzed pollen samples were placed on stubs

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using a nose hair-tipped needle (Zetter, 1989). The dry pollen samples were further dehydrated and cleaned in acidified 2, 2–dimethoxypropane (DMP) for 30 min, followed by critical point drying (Halbritter, 1998). This preparation technique was also performed for the study of the pollen morphology of rehydrated pollen grains, but before treatment with acidified DMP, the pollen samples were rehydrated with Agepon solution in distilled water (1:200) for 30 min. Other dry-untreated pollen samples from all *Gnetum* spp. were also investigated under SEM in order to ensure that their obtained characters of dehydration were not caused by sample preparation techniques. All pollen samples on SEM stubs were sputter-coated with gold using an SPI-module sputter coater (with SPI supplies) and then examined and photographed using an FEI Quanta 400 SEM. The characters studied were the pollen size, the type of microechinus tip, the echinus height and base width, the distance between microechini, the number of microechini per 25 μm^2 surface area and the number of microechini in each inflated region. The FEI XT microscope control software was used to measure the parameters mentioned. Twenty pollen grains were examined for each species in each treatment. However, measurements were not made in the study of the dry-untreated pollen samples. Pollen morphology was described using the terminology published in Punt *et al.* (2007) and Hesse *et al.* (2009).

RESULTS

The results are shown in evolutionary order, referring to the phylogenetic information studies of Hou *et al.* (2015) and Won & Renner (2006). In the present study, we examined pollen from eight *Gnetum* taxa found in Thailand: one arborescent Asian taxa (*Gnetum gnemon* var. *tenerum* from clade G) and seven lianoid Asian taxa: *G. montanum* from clade O, or the *hainanense* clade; *G. latifolium* var. *funiculare* and *G. leptostachyum* from clade M, or the *latifolium* clade; and *G. macrostachyum*, *G. tenuifolium*, *G. cuspidatum* and *G. microcarpum* from clade N, or the *cuspidatum* clade. The morphologies of their pollen are described as follows.

Gnetum gnemon var. *Tenerum*

Pollen are monads, subspheroidal, elliptic or ovate in the long axis view, circular or elliptic in the short axis view, long axis diameter 14–20 μm in LM, 11.13–16.78 μm in SEM (acetolyzed grains), 8.71–16.69 μm in SEM (non-acetolyzed grains), short axis diameter 12.0–17.5 μm in LM, 10.01–16.18 μm in SEM (acetolyzed grains), 8.27–13.11 μm in SEM (non-acetolyzed grains), inaperturate, exine 0.5–0.8 μm thick (LM), tectate, continuous tectum, microechinate in LM (showing inapparent microechini), microechinate with psilate or ridged surface between microechini in SEM, with a bluntly round tip. On some grains, there are remarkably raised exine regions surrounded by a ridge, with one to five microechini on each region. Microechini are small, 0.18–0.83 μm high and 0.28–0.64 μm in base diameter (acetolyzed grains), 0.16–0.44 μm high and 0.26–0.89 μm in base diameter (non-acetolyzed grains). There are 10–19 microechini per surface area of 25 μm^2 on acetolyzed grains and 11–22 microechini per surface area of 25 μm^2 on non-acetolyzed grains. The distances between microechini on acetolyzed grains range from 0.95 to 2.31 μm, whereas those on non-acetolyzed grains vary from 0.23 to 2.62 μm (Tables 1–3; Figs 1A, 2A–C, 4A–D).

Gnetum montanum

Pollen are monads, subspheroidal, elliptic or ovate in the long axis view, circular or elliptic in the short axis view, long axis diameter 12.0–17.5 μm in LM, 10.15–16.68 μm in SEM (acetolyzed grains), 10.15–16.71 μm in SEM (non-acetolyzed grains), short axis diameter 11.0–15.0 μm in LM, 8.34–15.85 μm in SEM (acetolyzed grains), 9.05–15.55 μm in SEM (non-acetolyzed grains), inaperturate, exine 0.6–0.9 μm thick (LM), tectate, continuous tectum, microechinate in LM, microechinate with psilate or ridged surface between microechini in SEM, with blunt-ended microechini. Some grains show slightly or considerably raised exine areas enclosed by a ridge. Numbers of microechini on each raised area vary from one to eight. Microechini are small, 0.33–0.69 μm high and 0.29–0.75 μm in base diameter (acetolyzed grains) and 0.21–0.81 μm high and 0.35–0.82 μm in base diameter (non-acetolyzed grains). In an area of 25 μm^2 on the tectum surface, ten to 18 and 11–16 microechini were found on acetolyzed grains and non-acetolyzed grains, respectively. The distances between microechini vary from 0.93 to 2.89 μm on acetolyzed grains and from 0.73 to 2.78 μm on non-acetolyzed grains (Tables 1–3; Figs 1B, 2D–F, 4E–H).

Gnetum latifolium var. *Funiculare*

Pollen are monads, subspheroidal, elliptic or ovate in the long axis view, circular or elliptic in the short axis view, long axis diameter 13–16 μm in LM, 9.84–16.36 μm in SEM (acetolyzed grains), 10.31–16.58 μm in SEM (non-acetolyzed grains), short axis diameter 10.0–15.0 μm in LM, 8.03–15.56 μm in SEM (acetolyzed grains), 8.54–14.20 μm in SEM (non-acetolyzed grains), inaperturate, exine 0.4–0.7 μm thick (LM), tectate, continuous tectum, microechinate in LM, microechinate with psilate or ridged surface between microechini in SEM, with a bluntly round tip. Some

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grains display distinct raised exine regions bordered by a ridge, with one to five microechini on each region. Microechini are 0.28–0.71 µm high and 0.29–0.79 µm in base diameter (acetolyzed grains), 0.32–0.64 µm high and 0.28–0.77 µm in base diameter (non-acetolyzed grains). There are nine to 16 microechini per surface area of 25 µm^2 on acetolyzed grains and ten to 32 on non-acetolyzed grains. The distances between microechini on acetolyzed grains are 0.55–2.58 µm, whereas those on non-acetolyzed grains are 0.40–2.56 µm (Tables 1–3; Figs 1C, 2G–I, 4I–L).

Gnetum leptostachyum

Pollen are monads, subspheroidal, elliptic or ovate in the long axis view, circular or elliptic in the short axis view, long axis diameter 12–16 µm in LM, 11.90–14.92 µm in SEM (acetolyzed grains), 11.46–14.10 µm in SEM (non-acetolyzed grains), short axis diameter 10–14 µm in LM, 10.03–13.81 µm in SEM (acetolyzed grains), 10.36–12.90 µm in SEM (non-acetolyzed grains), inaperturate, exine 0.4–0.8 µm thick (LM), tectate, continuous tectum, microechinate in LM, microechinate with psilate or ridged surface between microechini in SEM, with blunt-ended microechini. Some grains demonstrate considerably raised exine areas surrounded by a ridge, with one to seven microechini on each area. Sometimes, the base of a microechinus is swollen forming a dome-shaped microechinus. Microechini are small, 0.16–0.41 µm high and 0.21–0.67 µm in base diameter (acetolyzed grains), 0.22–0.75 µm high and 0.43–1.34 µm in base diameter (non-acetolyzed grains). In an area of 25 µm^2 on the tectum surface, ten to 15 and nine to 17 microechini were observed on acetolyzed grains and non-acetolyzed grains, respectively. The distances between microechini vary from 0.84 to 2.29 µm on acetolyzed grains and from 0.73 to 2.31 µm on non-acetolyzed grains (Tables 1–3; Figs 1D, 2J–L, 4M–P).

Gnetum macrostachyum

Pollen are monads, subspheroidal, elliptic or ovate in the long axis view, circular or elliptic in the short axis view, long axis diameter 12.5–16.5 µm in LM, 11.87–17.76 µm in SEM (acetolyzed grains), 12.06–14.30 µm in SEM (non-acetolyzed grains), short axis diameter 12.0–14.0 µm in LM, 11.23–16.77 µm in SEM (acetolyzed grains), 11.04–13.43 µm in SEM (non-acetolyzed grains), inaperturate, exine 0.5–0.8 µm thick (LM), tectate, continuous tectum, microechinate in LM, microechinate with psilate (not present in the acetolyzed sample) or ridged surface between microechini in SEM, with a bluntly round tip. On both acetolyzed and non-acetolyzed grains, dome-shaped microechini with blunt-ended tips were frequently

Table 1. Morphology of acetolyzed pollen grains from studied *Gnetum* species under LM. The average value and standard deviation are shown in parentheses.

Species	*G. gnemon*	*G. montanum*	*G. latifolium*	*G. leptostachyum*	*G. macrostachyum*	*G. tenuifolium*	*G. cuspidatum*	*G. microcarpum*
Pollen size (µm)	12.0–17.5 × 14.0–20.0 (14.6 ± 1.6 × 16.5 ± 1.5)	11.0–15.0 × 12.0–17.5 (13.5 ± 1.4 × 15.3 ± 1.7)	10.0–15.0 × 13.0–16.0 (13.3 ± 1.7 × 14.6 ± 0.7)	10.0–14.0 × 12.0–16.0 (12.2 ± 1.0 × 13.7 ± 0.9)	12.0–14.0 × 12.5–16.5 (12.6 ± 0.7 × 14.8 ± 1.2)	11.0–15.0 × 12.0–17.5 (12.9 ± 1.4 × 14.5 ± 1.3)	11.0–14.0 × 12.0–16.0 (12.8 ± 0.9 × 14.3 ± 0.9)	11.0–15.4 × 12.4–19.2 (12.5 ± 1.1 × 14.5 ± 1.8)
Exine thickness[a] (µm)	0.5–0.8 (0.6 ± 0.1)	0.6–0.9 (0.7 ± 0.1)	0.4–0.7 (0.6 ± 0.1)	0.4–0.8 (0.6 ± 0.1)	0.5–0.8 (0.6 ± 0.1)	0.4–0.8 (0.6 ± 0.1)	0.5–0.8 (0.7 ± 0.1)	0.5–0.6 (0.5 ± 0.1)

[a]Exine thickness excluding the echinus height

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Table 2. Morphology of acetolyzed pollen grains from studied Gnetum species under SEM ([a]n = 11; [b]n = 7). The average value and standard deviation are shown in parentheses.

Species	*G. gnemon*	*G. montanum*	*G. latifolium*	*G. leptostachyum*	*G. macrostachyum*	*G. tenuifolium*	*G. cuspidatum*	*G. microcarpum*
Acetolyzed pollen grains showing apparent ridge and raised exine region (SEM)								
Pollen size (µm)	10.01–12.53 × 11.13–13.74 (11.73 ± 0.84 × 12.61 ± 0.70)	8.58–15–85 × 10.15–16.48 (12.18 ± 2.79 × 13.57 ± 2.06)	8.03–15.14 × 9.84–15.89 (11.22 ± 1.98 × 12.42 ± 1.60)	10.98–13.81 × 11.90–14.83 (11.84 ± 077 × 13.00 ± 0.79)	11.23–16.77 × 11.87–17.76 (13.31 ± 1.47 × 13.96 ± 1.53)	11.33–15.11 × 11.82–16.42 (12.85 ± 1.28 × 13.99 ± 1.95)	11.20–15.62 × 11.62–16.03 (13.38 ± 1.82 × 13.69 ± 1.94)	11.20–13.62 × 12.34–14.47 (11.81 ± 0.87 × 13.35 ± 0.87[b])
Microechinus height (µm)	0.34–0.83 (0.59 ± 0.13)	0.33–0.69 (0.47 ± 0.09)	0.39–0.61 (0.47 ± .05)	0.16–0.36 (0.26 ± 0.06)	0.60–0.89 (0.72 ± 0.08)	0.22–0.78 (0.45 ± 0.16)	0.38–1.09 (0.71 ± 0.25)	0.23–0.36 (0.31 ± 0.04[b])
Microechinus base diameter (µm)	0.28–0.59 (0.47 ± 0.9)	0.29–0.75 (0.51 ± 0.09)	0.36–0.79 (0.48 ± 0.09)	0.21–0.67 (0.40 ± 0.13)	0.78–1.22 (0.93 ± 0.13)	0.38–1.34 (0.82 ± 0.28)	0.47–1.72 (0.99 ± 0.41)	0.34–0.58 (0.45 ± 0.08[b])
Distance between microechini (µm)	1.03–1.81 (1.40 ± 0.25)	0.93–2.11 (1.42 ± 0.29)	0.55–2.32 (1.42 ± 0.47)	0.84–2.15 (1.41 ± 0.24)	1.15–1.93 (1.47 ± 0.26)	0.78–2.33 (1.42 ± 0.40)	1.04–2.30 (1.45 ± 0.34)	1.15–2.18 (1.62 ± 0.39[b])
Number of microechini per 25 µm² (microechini/25 µm²)	14–19 (17.05 ± 1.50)	13–18 (15.55 ± 1.43)	10–16 (12.15 ± 1.46)	10–15 (12.75 ± 1.52)	8–11 (8.85 ± 0.81)	14–19 (16.00 ± 1.30)	9–14 (11.75 ± 1.25)	10–13 (11.14 ± 1.07)
Number of microechini in each raised exine region	1–5	1–8	1–3	1–6	1–3	2–11	1–3	1–3[b]
Acetolyzed pollen grains showing no (or inapparent) ridge and raised exine region (SEM)								
Pollen size (µm)	11.80–16.18 × 13.51–16.78 (13.94 ± 1.54 × 15.34 ± 0.95)	8.34–14.65 × 10.49–16.68 (11.66 ± 2.5 × 13.46 ± 2.10)	9.33–15.56 × 11.52–16.36 (12.11 ± 1.77 × 13.88 ± 1.67)	10.03–13.50 × 11.95–14.92 (11.82 ± 1.10 × 13.22 ± 0.82[a])	-	10.56–15.80 × 11.51–17.88 (12.91 ± 1.90 × 14.63 ± 2.53)	11.15–16.56 × 11.85–17.00 (13.30 ± 1.73 × 13.98 ± 1.80)	10.27–15.61 × 11.99–21.70 (12.60 ± 1.66 × 14.38 ± 3.41)
Microechinus height (µm)	0.18–0.42 (0.28 ± 0.07)	0.37–0.69 (0.54 ± 0.07)	0.28–0.71 (0.48 ± 0.09)	0.30–0.41 (0.35 ± 0.04[a])	-	0.15–0.78 (0.42 ± 0.19)	0.22–0.75 (0.44 ± 0.13)	0.16–0.49 (0.32 ± 0.09)
Microechinus base diameter (µm)	0.32–0.64 (0.45 ± 0.09)	0.46–0.75 (0.59 ± 0.07)	0.29–0.67 (0.51 ± 0.80)	0.38–0.67 (0.55 ± 0.08[a])	-	0.21–0.75 (0.45 ± 0.13)	0.38–0.76 (0.53 ± 0.12)	0.26–0.51 (0.40 ± 0.07)
Distance between microechini (µm)	0.95–2.31 (1.63 ± 0.36)	1.00–2.89 (1.71 ± 0.40)	1.12–2.58 (1.71 ± 0.38)	1.22–2.29 (1.73 ± 0.29[a])	-	1.29–2.68 (1.86 ± 0.40)	1.32–2.93 (1.92 ± 0.49)	1.28–2.10 (1.64 ± 0.19)
Number of microechini per 25 µm² (microechini/25 µm²)	10 -14 (12.45 ± 1.23)	10–15 (11.40 ± 1.27)	9–13 (10.35 ± 1.35)	10–13 (11.82 ± 0.75 [a])	-	6–12 (8.95 ± 1.76)	6–10 (8.20 ± 1.15)	7–13 (9.60 ± 1.67)

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Table 3. Morphology of non-acetolyzed pollen grains from studied Gnetum species under SEM ([a]n = 2; [b]n = 8). The average value and standard deviation are shown in parentheses.

Species	*G. gnemon*	*G. montanum*	*G. latifolium*	*G. leptostachyum*	*G. macrostachyum*	*G. tenuifolium*	*G. cuspidatum*	*G. microcarpum*
Dry-treated pollen grains showing apparent ridge and raised exine region (SEM)								
Pollen size (µm)	8.27–12.17 × 8.71–12.48 (9.91 ± 1.23 × 10.84 ± 1.18)	9.05–15.55 × 10.15–16.23 (12.27 ± 2.70 × 13.33 ± 2.07)	8.54–14.2 × 10.31–16.46 (11.07 ± 1.74 × 12.61 ± 1.91)	10.54–12.90 × 11.46–14.10 (11.74 ± 0.56 × 12.70 ± 0.60)	11.04–13.43 × 12.06–14.30 (11.90 ± 0.70 × 12.75 ± 0.61)	9.72–16.29 × 10.45–17.06 (13.06 ± 2.46 × 13.86 ± 2.55)	11.01–15.62 × 12.61–16.07 (13.34 ± 1.94 × 14.36 ± 1.39	9.70–12.51 × 10.88–12.68 (10.61 ± 0.85 × 11.68 ± 0.37)
Microechinus height (µm)								
	0.16–0.44 (0.31 ± 0.07)	0.57–0.81 (0.70 ± 0.08)	0.36–0.61 (0.52 ± 0.08)	0.25–0.75 (0.46 ± 0.11)	0.44–0.83 (0.63 ± 0.11)	0.17–0.75 (0.42 ± 0.17)	0.51–1.05 (0.77 ± 0.16)	0.47–0.89 (0.61 ± 0.13)
Microechinus base diameter (µm)								
	0.26–0.89 (0.51 ± 0.14)	0.44–0.82 (0.62 ± 0.11)	0.28–0.60 (0.46 ± 0.07)	0.47–1.34 (0.79 ± 0.20)	0.70–1.26 (0.98 ± 0.16)	0.34–1.20 (0.71 ± 0.32)	0.39–1.71 (1.02 ± 0.35)	0.71–1.31 (0.88 ± 0.14)
Distance between microechini (µm)								
	0.23–1.04 (0.70 ± 0.24)	0.73–2.10 (1.26 ± 0.33)	0.40–2.04 (1.14 ± 0.61)	0.73–1.80 (1.26 ± 0.25)	0.94–1.81 (1.18 ± 0.23)	0.64–1.60 (1.28 ± 0.26)	1.06–1.88 (1.44 ± 0.23)	0.78–1.56 (1.09 ± 0.23)
Number of microechini per 25 µm² (microechini/25 µm²)								
	15–22 (19.45 ± 2.16)	12–16 (13.75 ± 1.25)	10–32 (13.65 ± 5.46)	12–17 (15.00 ± 1.26)	10–14 (12.25 ± 1.07)	12–20 (16.25 ± 2.27)	8–14 (11.50 ± 1.50)	13–21 (17.35 ± 1.78)
Number of microechini in each raised exine region								
	1–4	1–5	1–5	1–7	1–2	1–4	1–4	1–3
Rehydrated pollen grains showing no (or inapparent) ridge and raised exine region (SEM)								
Pollen size (µm)	10.36–13.11 × 13.14–16.69 (12.17 ± 0.69 × 14.08 ± 0.89)	9.44–15.21 × 10.98–16.71 (12.75 ± 2.00 × 14.32 ± 1.94)	9.15-12-93 × 11.51–16.58 (11.13 ± 1.12 × 12.88 ± 0.96)	10.36–12.37 × 11.60–13.61 (11.43 ± 0.57 × 12.93 ± 0.65)	12.22–13.03 × 12.72–13.27 (12.63 ± 0.56 × 13.00 ± 0.39[a])	10.35–16.20 × 11.71–17.85 (12.79 ± 1.98 × 14.59 ± 2.51)	11.96–16.56 × 12.31–16.88 (14.24 ± 1.98 × 14.68 ± 1.97[b])	11.51–18.85 × 12.01–21.90 (13.99 ± 2.59 × 15.85 ± 3.84)
Microechinus height (µm)								
	0.19–0.42 (0.32 ± 0.06)	0.21–0.48 (0.35 ± 0.08)	0.32–0.64 (0.52 ± 0.09)	0.22–0.45 (0.35 ± 0.06)	0.47–0.52 (0.50 ± 0.04[a])	0.14–0.41 (0.26 ± 0.07)	0.33–0.72 (0.50 ± 0.13[b])	0.15–0.49 (0.34 ± 0.08)
Microechinus base diameter (µm)								
	0.29–0.63 (0.45 ± 0.09)	0.35–0.62 (0.47 ± 0.08)	0.38–0.77 (0.55 ± 0.11)	0.43–0.83 (0.67 ± 0.11)	0.88–0.91 (0.90 ± 0.02[a])	0.2–0.7 (0.47 ± 0.13)	0.48–1.11 (0.75 ± 0.21[b])	0.19–0.56 (0.40 ± 0.09)
Distance between microechini (µm)								
	0.85–2.62 (1.59 ± 0.53)	1.09–2.78 (1.87 ± 0.43)	1.11–2.56 (1.78 ± 0.43)	1.07–2.31 (1.61 ± 0.28)	2.07–2.69 (2.38 ± 0.44[a])	1.38–2.50 (1.84 ± 0.32)	1.40–2.29 (1.90 ± 0.33[b])	1.19–2.11 (1.62 ± 0.25)
Number of microechini per 25 µm² (microechini/25 µm²)								
	11–16 (13.40 ± 1.14)	11–14 (12.70 ± 0.98)	10 -14 (11.50 ± 1.00)	9–13 (10.85 ± 1.09)	10–11 (10.50 ± 0.71[a])	7–12 (8.60 ± 1.19)	8–12 (9.87 ± 1.55)	11–15 (12.85 ± 1.04)

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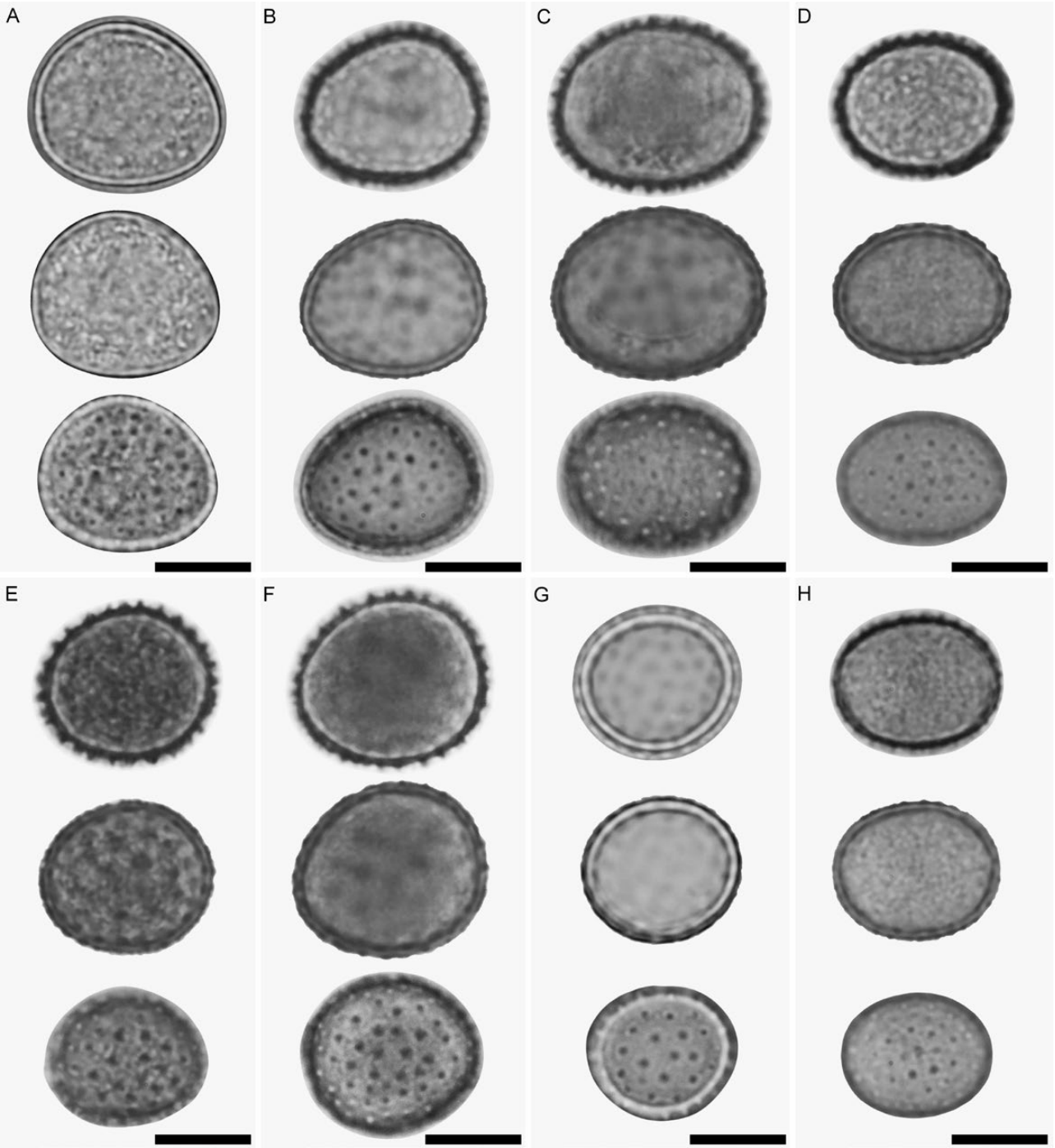

Figure 1. Light micrographs (LM) of *Gnetum* pollen species in Thailand (A-H). A, *Gnetum gnemon*. B, *Gnetum montanum*. C, *Gnetum latifolium*. D, *Gnetum leptostachyum*. E, *Gnetum macrostachyum*. F, *Gnetum tenuifolium*. G, *Gnetum cuspidatum*. H, *Gnetum microcarpum*. Scale bars: 10 µm (A–H).

observed adjacent to each other, a ridge surrounding one or a few of them, forming a network-like pattern on the pollen wall. Pollen grains with a crumpled tectum surface could be observed in the acetolyzed pollen samples. Microechini are small, 0.60–0.89 µm high and 0.78–1.22 µm in base diameter (acetolyzed grains), 0.44–0.83 µm high and 0.70–1.26 µm in base diameter (non-acetolyzed grains). There are eight to 11 microechini per surface area of 25 µm^2 on acetolyzed grains and ten to 14 on non-acetolyzed grains. The

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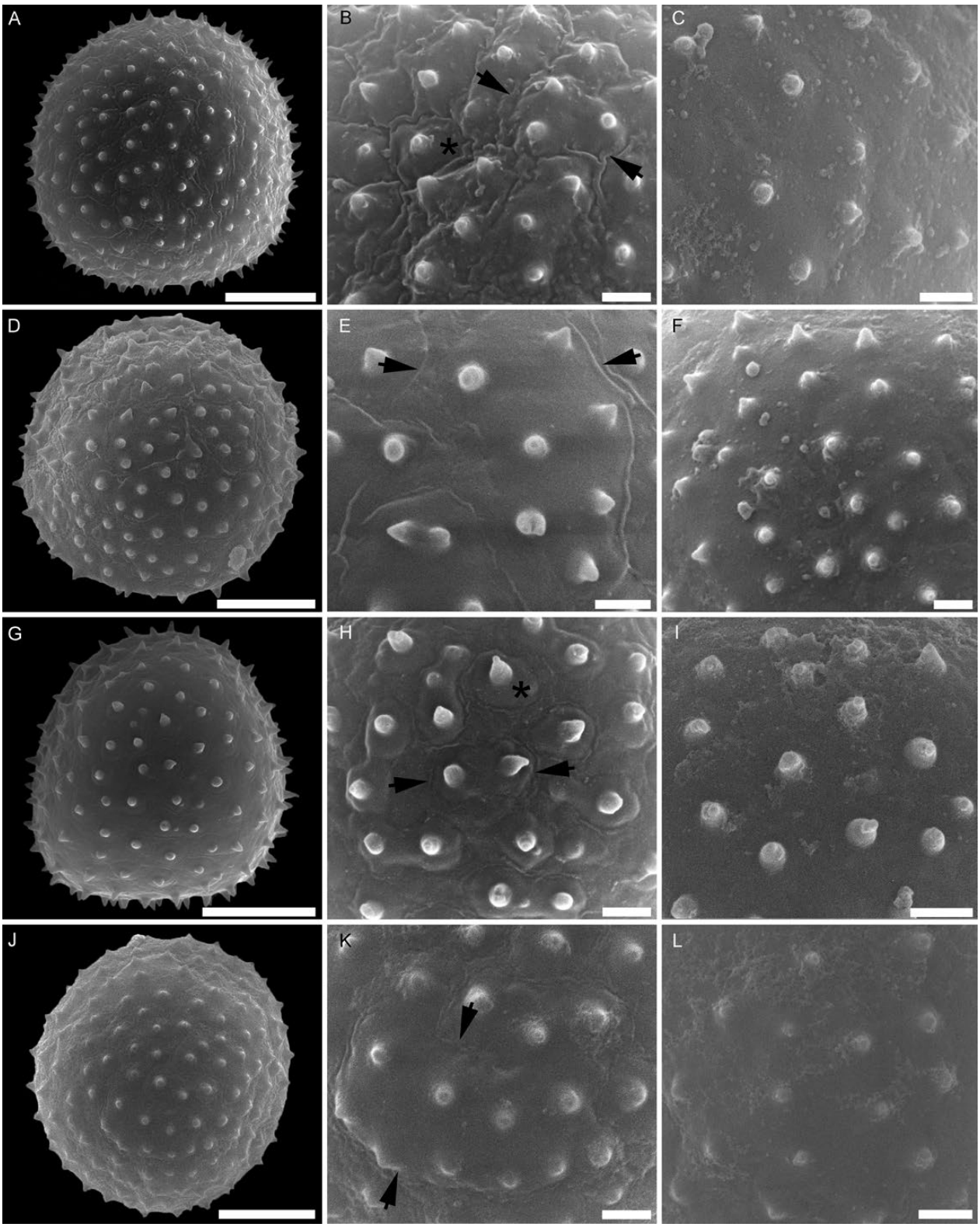

Figure 2. Scanning electron micrographs (SEM) of acetolyzed pollen grains from *Gnetum* species in Thailand showing the variation in sculptures (A–L). A–C, *Gnetum gnemon*. D–F, *Gnetum montanum*. G–I, *Gnetum latifolium*. J–L, *Gnetum leptostachyum* (A, D, G and J, general view; B, E, H and K, echinate with apparent ridge and raised exine region; C, F, I and L, echinate with no, or inapparent, ridge and raised exine region). Arrows indicate a raised exine region surrounded by a ridge. An asterisk indicates a dome-shaped microechinus. Scale bars: 10 µm (A, D, G and J); 1 µm (B, C, E, F, H, I, K and L).

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distances between microechini on acetolyzed grains are 1.15–1.93 µm, whereas those on non-acetolyzed grains are 0.94–2.69 µm (Tables 1–3; Figs 1E, 3A–C, 5A–D).

Gnetum tenuifolium

Pollen are monads, subspheroidal, elliptic or ovate in the long axis view, circular or elliptic in the short axis view, long axis diameter 12.0–17.5 µm in LM, 11.51–17.88 µm in SEM (acetolyzed grains), 10.45–17.85 µm in SEM (non-acetolyzed grains), short axis diameter 11–15 µm in LM, 10.56–15.80 µm in SEM (acetolyzed grains), 9.72–16.29 µm in SEM (non-acetolyzed grains), inaperturate, exine 0.4–0.8 µm thick (LM), tectate, continuous tectum, microechinate in LM, microechinate with psilate (rare) or ridged surface between microechini in SEM, with blunt-ended microechini. Pollen grains with distinct and large raised exine regions surrounded by a ridge, with two to 11 microechini on each region, were frequently recognized. A dome-shaped microechinus with blunt apex was also observed. Microechini are 0.15–0.78 µm high and 0.21–1.34 µm in base diameter (acetolyzed grains), 0.14–0.75 µm high and 0.20–1.20 µm in base diameter (non-acetolyzed grains). In an area of 25 µm^2 on the tectum surface, six to 19 and seven to 20 microechini were recognized on acetolyzed grains and non-acetolyzed grains, respectively. The distances between microechini vary from 0.78 to 2.68 µm on acetolyzed grains and from 0.64 to 2.50 µm on non-acetolyzed grains (Tables 1–3; Figs 1F, 3D–F, 5E–H).

Gnetum cuspidatum

Pollen are monads, subspheroidal, elliptic or ovate in the long axis view, circular or elliptic in the short axis view, long axis diameter 12–16 µm in LM, 11.62–17.00 µm in SEM (acetolyzed grains), 12.31–16.88 µm in SEM (non-acetolyzed grains), short axis diameter 11–14 µm in LM, 11.15–16.62 µm in SEM (acetolyzed grains), 11.01–16.56 µm in SEM (non-acetolyzed grains), inaperturate, exine 0.5–0.8 µm thick (LM), tectate, continuous tectum, microechinate in LM, microechinate with psilate or ridged surface between microechini in SEM, with a bluntly round tip. Some grains demonstrate slightly or considerably raised exine areas surrounded by the ridge, with one to four microechini on each area. On dry pollen samples, dome-shaped microechini appear to be more adjacent to each other than those on acetolyzed samples. Microechini are small, 0.22–1.09 µm high and 0.38–1.72 µm in base diameter (acetolyzed grains), 0.33–1.05 µm high and 0.39–1.71 µm in base diameter (non-acetolyzed grains). There are six to 14 microechini per surface area of 25 µm^2 on acetolyzed grains and eight to 14 microechini per surface area of 25 µm^2 on non-acetolyzed grains. The distances between microechini vary from 1.04 to 2.93 µm on acetolyzed grains and from 1.06 to 2.29 on non-acetolyzed grains (Tables 1–3; Figs 1G, 3G–I, 5I–L).

Gnetum microcarpum

Pollen are monads, subspheroidal, elliptic or ovate in the long axis view, circular or elliptic in the short axis view, long axis diameter 12.4–19.2 µm in LM, 11.99–21.70 µm in SEM (acetolyzed grains), 10.88–21.90 µm in SEM (non-acetolyzed grains), short axis diameter 11.0–15.4 µm in LM, 10.27–15.61 µm in SEM (acetolyzed grains), 9.70–18.85 µm in SEM (non-acetolyzed grains), inaperturate, exine 0.5–0.6 µm thick (LM), tectate, microechinate in LM, microechinate with a psilate or (rarely) ridged surface between microechini in SEM, with a bluntly round tip. Some grains demonstrate slightly raised exine areas enclosed by a ridge, with one to three microechini on each area. Microechini are small, 0.16–0.49 µm high and 0.26–0.58 µm in base diameter (acetolyzed grains), 0.15–0.89 µm high and 0.19–1.31 µm in base diameter (non-acetolyzed grains). In an area of 25 µm^2 on the tectum surface, seven to 13 and 11–21 microechini were observed on acetolyzed grains and non-acetolyzed grains, respectively. The distances between microechini vary from 1.15 to 2.18 µm on acetolyzed grains and from 0.78 to 2.11 on non-acetolyzed grains (Tables 1–3; Figs 1H, 3J–L, 5M–P).

Raised exine regions enclosed by ridges were found in both dry-treated and dry-untreated pollen grains of all *Gnetum* spp. (Figs 4A, C, E, G, I, K, M and O, 5A, C, E, G, I, K, M and O and 6A–P).

DISCUSSION

Pollen morphology of *Gnetum* species

Pollen from all *Gnetum* spp. studied were monads, subspheroidal, elliptic or ovate in the long axis view, circular or elliptic in the short axis view, inaperturate, tectate with a continuous tectum, and microechinate with psilate or ridged surface between microechini. Their morphology is generally similar to those of Asian *Gnetum* spp. described in previous studies, including *G. cleistostachyum* C.Y.Cheng, *G. costatum* K.Schum., *G. cuspidatum*, *G. gnemon*, *G. hainanense* C.Y.Cheng, *G. latifolium* var. *funiculare* (as *G. funiculare* Blume), *G. leptostachyum*, *G. luofuense* C.Y.Cheng, *G. macrostachyum*, *G. montanum*, *G. neglectum* Blume, *G. parvifolium* (Warb.) Cheng (=*G. indicum* Merr.) and *G. pendulum* C.Y.Cheng (Erdtman 1954, 1965; Hesse,

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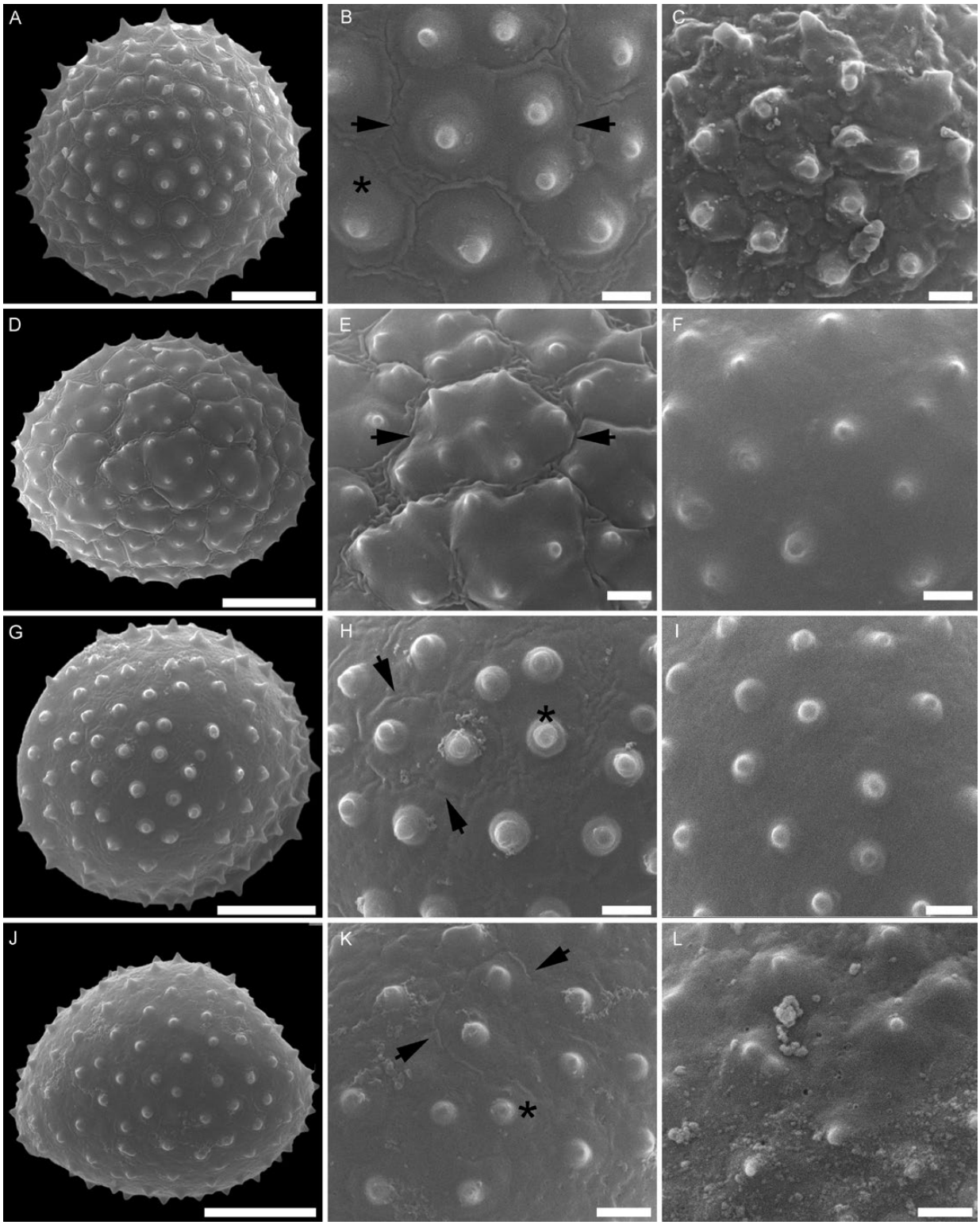

Figure 3. Scanning electron micrographs (SEM) of acetolyzed pollen grains from *Gnetum* species in Thailand showing the variation in sculptures (A–L). A–C, *Gnetum macrostachyum*. D–F, *Gnetum tenuifolium*. G–I, *Gnetum cuspidatum*. J–L, *Gnetum microcapum* (A, D, G and J, general view; B, E, H and K, echinate with apparent ridge and raised exine region; C, crumpled pollen grain; F, I and L, echinate with no, or inapparent, ridge and raised exine region). Arrows indicate a raised exine region surrounded by a ridge. An asterisk indicates a dome-shaped microechinus. Scale bars: 10 µm (A, D, G and J); 1 µm (B, C, E, F, H, I, K and L).

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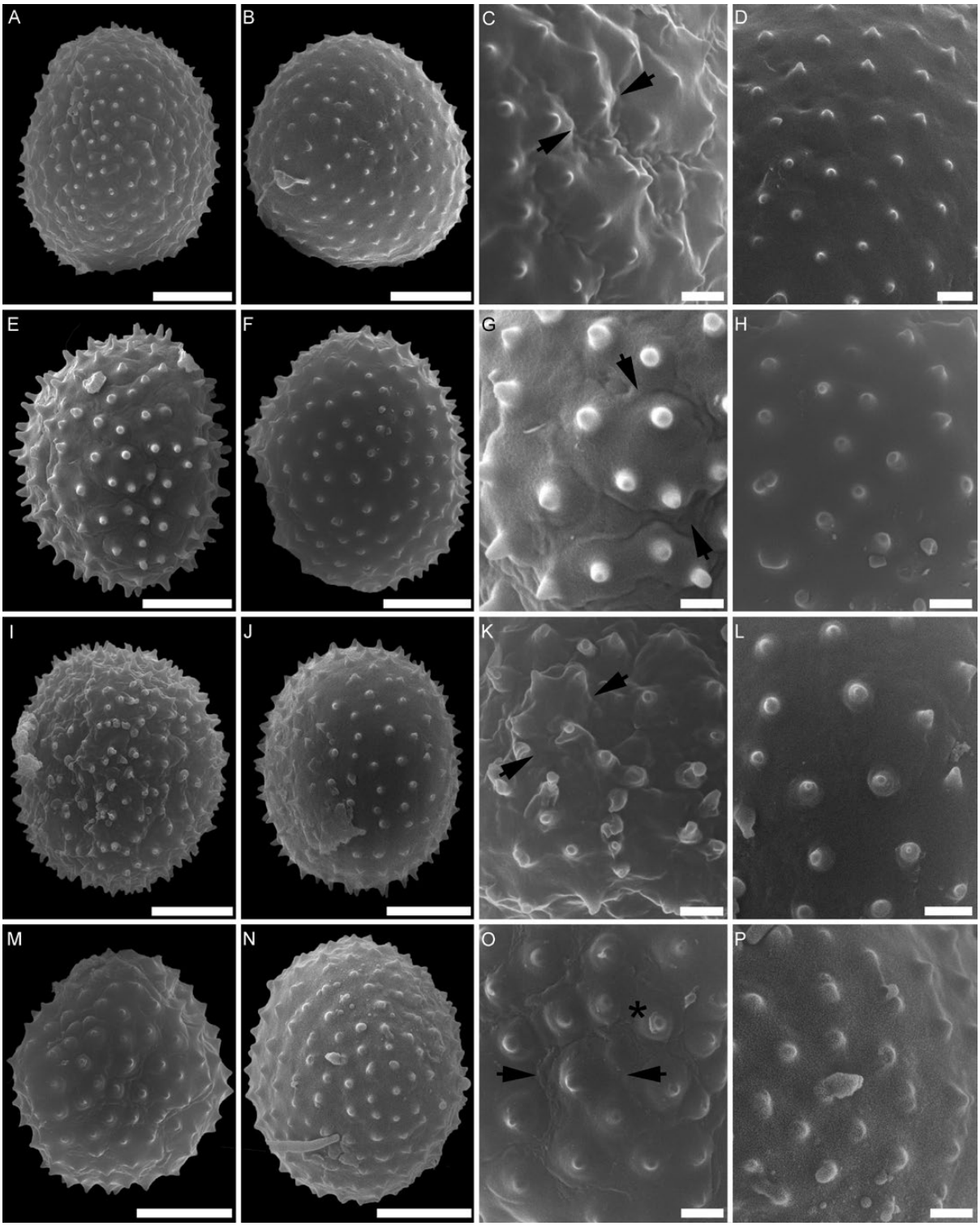

Figure 4. Scanning electron micrographs (SEM) of non-acetolyzed pollen grains from *Gnetum* species in Thailand showing the variation in sculptures (A–P). A–D, *Gnetum gnemon*. E–H, *Gnetum montanum*. I–L, *Gnetum latifolium*. M–P, *Gnetum leptostachyum* (A, E, I and M, dry-treated samples, general view; B, F, J and N, rehydrated samples, general view; C, G and O, dry-treated samples, echinate with apparent ridge and raised exine region; K, dry-treated sample, crumpled pollen grain; D, H, L and P, rehydrated samples, echinate with no, or inapparent, ridge and raised exine region). Arrows indicate a raised exine region surrounded by a ridge. An asterisk indicates a dome–shaped microechinus. Scale bars: 10 µm (A, B, E, F, I, J, M and N); 1 µm (C, D, G, H, K, L, O and P).

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1980; Gillespie & Nowicke, 1994; Osborn, 2000; Yao *et al.*, 2004; Tekleva, 2016). Even though, based on plant morphology, *G. gnemon* and *G. costatum* are placed in a different section to other Asian taxa (Griffith, 1859; Markgraf, 1929; Price, 1996), their pollen present a continuous tectum with microechini, which is similar to those of other Asian *Gnetum* spp. (Figs 1–6; Gillespie & Nowicke, 1994, plate I, fig. 1). Thus, this character supports the molecular phylogenetic information from studies by Won & Renner (2006), Hou *et al.* (2015) and Hou *et al.* (2016), revealing Asian *Gnetum* taxa to be monophyletic. In other words, it indicates that *Gnetum* section *Gnetum* subsection *Eugnemones* is more closely related to *Gnetum* section *Scandentia* than the other two groups in its own section [subsections *Araeognemones* (American taxa) and *Micrognemones* (African taxa)], which show a discontinuous tectum forming islands with numerous microechini (Tekleva, 2016, fig. 1a, b; Gillespie & Nowicke, 1994, plate II, figs 7–12). From the different tectal ornamentations observed under LM, Erdtman (1954, 1965) proposed three different groups of *Gnetum* pollen with non-overlapping geographical distributions, Asian, American and African. This proposal provides support for the three major phylogenetic clusters put forward by Hou *et al.* (2015). However, Gillespie & Nowicke (1994) showed that, in SEM, the tectal ornamentation of *Gnetum* pollen from the examined American species was not pilate as shown in those previous LM studies, being more similar to those of African taxa, plate II, figs 7–12). According to Hou *et al.* (2015), *Gnetum* diverged in the Late Cretaceous when the breakup of Gondwana occurred and the Asian crown group was dated to the Cretaceous-Palaeogene (K-Pg) boundary. These findings disagree with the hypothesis proposed by Markgraf (1929) that arborescent *Gnetum* taxa appeared to be an earlier diverged ancestor of other species in the genus. The discovery of the first fossil male strobiles with an affinity to *Gnetum*, *Khitania columnispicata* Guo, Sha, Bian & Qiu, from the Early Cretaceous in the Yixian Formation, Liaoning, China also confirms the results from the molecular study (Guo *et al.*, 2009). However, fossil palynomorphs comparable to pollen of extant *Gnetum* taxa have not yet been reported (Friis *et al.*, 2011). Such disappearances from the geological record are possibly associated with their thin ectexine layer (Tekleva, 2016).

Pollen of all examined *Gnetum* spp. displayed morphological plasticity. Some grains exhibited a psilate surface between microechini, whereas others showed slightly or considerably raised exine regions surrounded by various patterns of ridge-like folds, with one or more microechini on each region. Other studies on Asian *Gnetum* pollen also found similar raised regions and folding (Gillespie & Nowicke, 1994; Osborn, 2000; Yao *et al.*, 2004; Tekleva, 2016) but only Gillespie & Nowicke (1994; plate I, figs 2–3) reported the variation in tectum ornamentation of *Gnetum* pollen and also pointed out that the different sculptures observed could be caused by the sample preparation. However, two months after the first examination was carried out, we re-examined the acetolyzed grains with psilate surfaces between the microechini and found that the exines were slightly raised and ridged, and that the same characters that indicate the dehydrated state of *Gnetum* pollen grains were present in the dry-treated samples of all species as well as in the dry-untreated pollen grains. Thus, it can be concluded that, rather than sample preparation techniques, such morphological plasticity could be associated with harmomegathy, which we will discuss in the next section.

Even though the exine ultrastructures of pollen from *Ephedra* L., *Gnetum* and *Welwitschia* Hook.f. are similar, being described as granular infratectum, lamellate endexine, tectum protruding outwards, their exine ornamentations are different, being plicate in *Ephedra* and *Welwitschia* and microechinate in *Gnetum* (Gullvåg, 1966; Hesse, 1980, 1984; Osborn, Taylor & de Lima, 1993; Gillespie & Nowicke, 1994; Ickert-Bond, Skvarla & Chissoe, 2003; Yao *et al.*, 2004; Bolinder, Niklas & Rydin, 2015). Since this ornamentation of *Gnetum* pollen is unique among gymnosperms (Gillespie & Nowicke, 1994), it appears to be a potential synapomorphy for this plant group. Tekleva (2016) stated that in Asian *Gnetum* spp., raised exine areas with microechini are equivalent to the plate-like exine wall of *G. africanum* Welw. and the plicae of *Ephedra* and *Welwitschia*. *Gnetum* have been considered entomophilous, although various anemophilous syndromes have been recognized (Kato, Inoue & Nagamitsu, 1995; Kato, 1996; Momose, *et al.* 1998; Kato *et al.*, 2008; Gong *et al.*, 2015). It is possible that the adaptation of the *Gnetum* lineage reached an entomophilous pollination mode along a different line than *Ephedra* and *Welwitschia* (Tekleva, 2016). Jörgensen & Rydin (2015) proposed that the loss of sterile ovules in African *Gnetum* is probably linked to wind pollination. However, the correlation remains unclear between wind pollination and a discontinuous tectum as the pollination biology of American species that also produce pollen with a discontinuous tectum has not yet been investigated.

A cup-like depression was found in some grains under LM and SEM in our study but without any indication of its being an exit for a pollen tube. This depression was interpreted as a leptoma (pore-like area) by Erdtman (1965) and Kuprianova (1983). Gillespie & Nowicke (1994), Yao *et al.*, (2004) and Tekleva (2016) also found, under SEM and TEM, no

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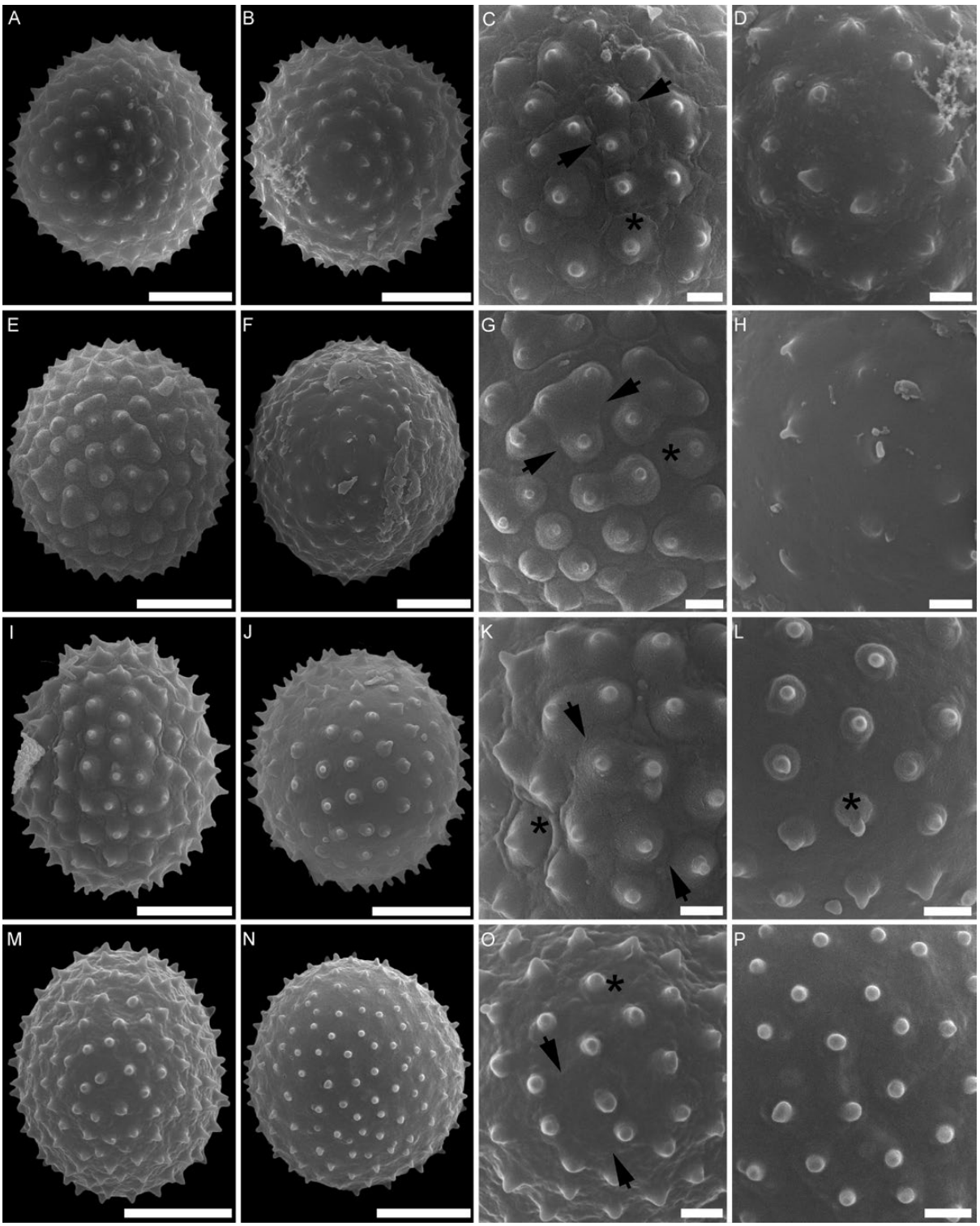

Figure 5. Scanning electron micrographs (SEM) of non-acetolyzed pollen grains from *Gnetum* species in Thailand showing the variation in sculptures (A–P). A–D, *Gnetum macrostachyum*. E–H, *Gnetum tenuifolium*. I–L, *Gnetum cuspidatum*. M–P, *Gnetum microcarpum* (A, E, I and M, dry-treated samples, general view; B, F, J and N, rehydrated samples, general view; C, G, K and O, dry-treated samples, echinate with apparent ridge and raised exine region; D, H, L and P, rehydrated samples, echinate with no, or inapparent, ridge and raised exine region). Arrows indicate a raised exine region surrounded by a ridge. An asterisk indicates a dome-shaped microechinus. Scale bars: 10 µm (A, B, E, F, I, J, M and N); 1 µm (C, D, G, H, K, L, O and P).

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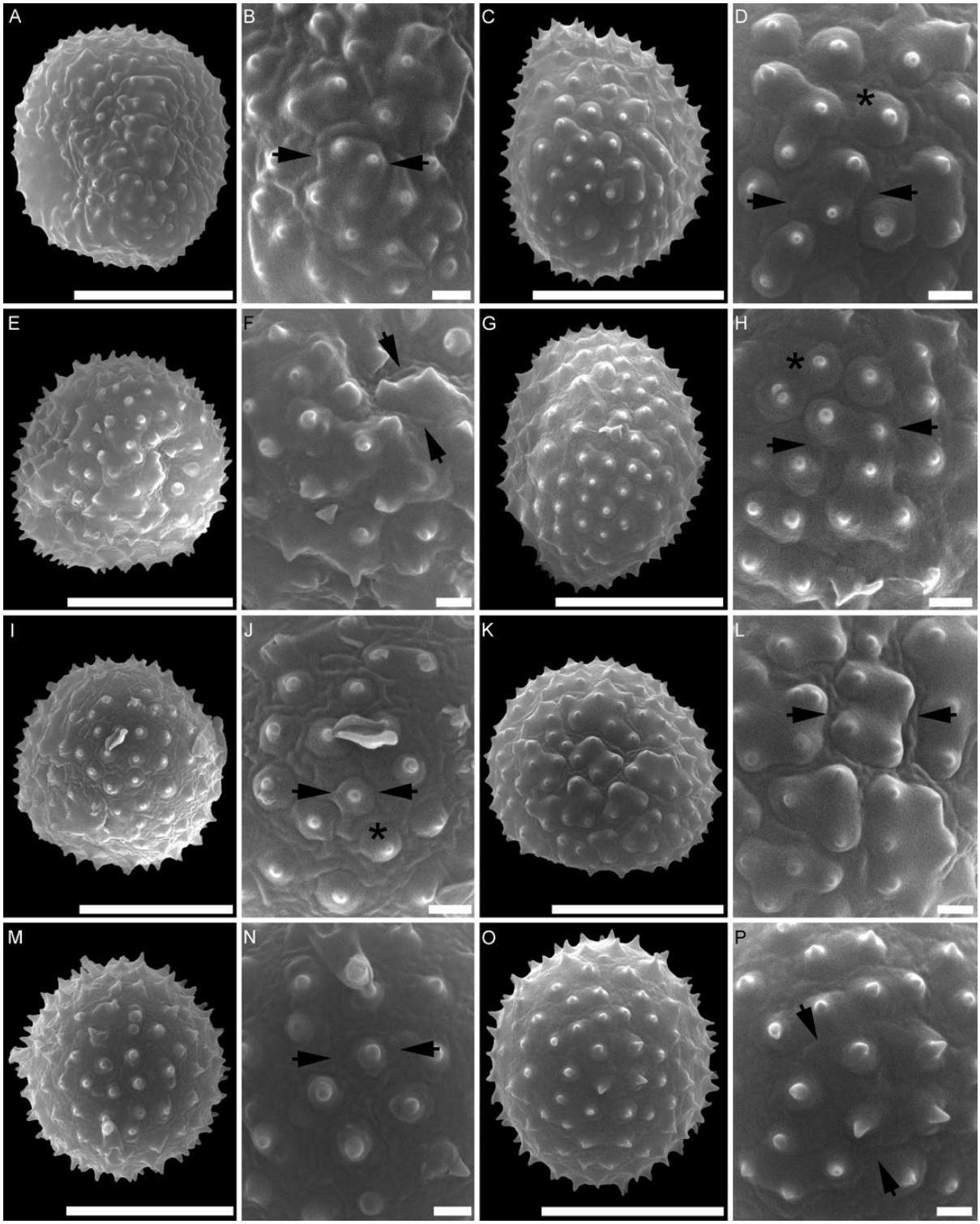

Figure 6. Scanning electron micrographs (SEM) of dry-untreated pollen grains from *Gnetum* species in Thailand showing the typical characters of the dehydrated state (A–P). A–B, *Gnetum gnemon*. C–D, *Gnetum montanum*. E–F, *Gnetum latifolium*. G–H, *Gnetum leptostachyum*. I–J, *Gnetum macrostachyum*. K–L, *Gnetum tenuifolium*. M–N, *Gnetum cuspidatum*. O–P, *Gnetum microcarpum* (A, C, E, G, I, K, M and O, general view; B, D, F, H, J, L, N and P, echinate with apparent ridge and raised exine region. Arrows indicate a raised exine region surrounded by a ridge. An asterisk indicates a dome-shaped microechinus. Scale bars: 10 µm (A, C, E, G, I, K, M and O); 1 µm (B, D, F, H, J, L, N and P).

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evidence of thinning on the exine wall to indicate the existence of an apertural region in *Gnetum* pollen.

In this study, a considerable overlap in qualitative and quantitative morphological characters exists among species and also among samples of the same species prepared by different techniques. Therefore, it is not possible to classify Thai *Gnetum* pollen into species, particularly because the pollen has a psilate surface between microechini. However, some differences could be noticed. Pollen from *G. gnemon* var. *tenerum* in this study shows the highest average values of grain size (in LM), and number of microechini within a tectum surface area of 25 μm^2 and the lowest average values of distance between microechini. These results imply that pollen of this taxon are large with numerous microechini. Moreover, under LM, their microechini are less prominent than those of others. The study by Osborn (2000), based on SEM, described *G. gnemon* pollen as spheroidal in shape, whereas they appeared subspheroidal in our LM investigation. Since a coverslip was not used in our study, the pollen could be turned over with a hair-tipped needle. As a result, we could obtain the correct position for measuring and photographing the pollen under LM. Due to the impossibility of moving pollen grains during SEM analysis, the dimensions of pollen measured under SEM differ from their dimensions measured under LM.

Yao *et al.* (2004) separated *Gnetum* pollen into two types according to their spinule (echinus) characters: one with inflated bases in *G. parvifolium* and *G. cleistostachyum*, and one without inflated bases in *G. montanum*, *G. luofuense*, *G. pendulum* and *G. hainanens*. However, in our analysis, both types were found in all examined pollen species due to their harmomegathic mechanism, although some different patterns of inflated areas could be recognized. In this study, the inflated bases of the microechini of pollen from *G. gnemon* var. *tenerum*, *G. latifolium* var. *funiculare*, *G. leptostachyum*, *G. montanum* and *G. tenuifolium* tended to merge with the bases of adjacent microechini, forming a raised exine region enclosed by a ridge with more than two microechini on each area (Figs 2B, E, H and K, 3E, 4C, G, K and O, 5G, 6A–H, K and L). This morphology was also found in pollen from *G. gnemon*, *G. latifolium* var. *funiculare*, *G. leptostachyum* and *G. montanum* studied by Gillespie & Nowicke (1994, plate I, figs 4–5), Halbritter (2015) and Tekleva (2016, fig. 1c–d). In the present study, numerous microechini on a large, merged, inflated base comprised a morphological character found only in pollen grains from *G. tenuifolium* (Figs 3D and E, 5E and G, 6K and L). The same morphological feature was also observed in pollen samples from different plants of this taxon. However, a single dome-shaped microechinus with an isolated, inflated base on a raised area were also found in all four species, more especially in dry pollen samples (both treated and untreated) from *G. leptostachyum* and *G. tenuifolium*.

The dome-shaped microechinus in our study, was more frequently observed in pollen from *G. macrostachyum*, *G. cuspidatum* and *G. microcarpum* (Figs 3B, H and K, 5C, K and O, 6I–J and M–P). One or a few dome-shaped microechini of *G. macrostachyum* pollen were bordered by a prominent ridge, resulting in the formation of a network-like sculpture on the pollen wall. These characters are similar to those of *G. macrostachyum* shown in Tekleva (2016, fig. 1 h–i). However, similar network-like patterns of ridges were also found in pollen from *G. indicum* (Tekleva, 2016, fig. 1e–f) and *G. latifolium* (Gillespie & Nowicke, 1994, plate I, fig. 3) but without the inflated base of the microechinus. Likewise, in this present study, the base of microechini of acetolyzed pollen of *G. cuspidatum* and *G. microcarpum*, and of the dry pollen samples of the latter species, appear slightly inflated and ridged. This was also recognized in *G. cuspidatum* pollen examined by Gillespie & Nowicke (1994, plate I, fig. 6).

Even though the morphologies of both dry-treated and dry-untreated pollen samples appeared to be similar, differences in some specific characters were observed. In *G. tenuifolium*, for example, the islands on the dry-treated pollen tended to be more distant from each other than they were on the dry-untreated pollen, even though both samples were taken from the same plant material (Figs 5E, G, 6K, L). This suggested that the variation in the characters of the dehydrated pollen could have been produced by the sample preparation methods and by the age of the specimens. Therefore, it should be ensured that the same technique is employed when a comparative study is made. However, different characters of dry pollen from *G. latifolium* were recognized even when the same preparation procedure was used (Fig. 6E, F; Tekleva, 2016, fig. 1c–d). Thus, it should be emphasized that intraspecific variation should also be taken into account when investigating pollen of this plant group.

THE HARMOMEGATHIC CHARACTER

In nature, pollen pass through many developmental stages that differ in relation to their hydrated and dehydrated states. These stages include the microsporogenesis phase (microspore or pollen immersed in locular fluid), the anther dehydration phase (the locular fluid disappears), the presentation phase (the anther opens), the dispersal phase (pollen are exposed to the environment) and the pollen-stigma interaction phase (pollen rehydrate on the stigma) (Firon, Nepi & Pacini, 2012). During these phases,

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the exine elasticity of pollen adapts to facilitate harmomegathy (Mohl, 1834). In this study, the average values of pollen size and distance between microechini of non-acetolyzed grains with psilate surfaces between microechini (rehydrated pollen) tended to be higher than those of non-acetolyzed grains with apparent ridges and raised exine areas (dry pollen), whereas the latter grains had higher average values for the microechinus height and base diameter and for the number of microechini in the defined area. The same tendency also existed in the acetolyzed pollen samples. In other words, pollen grains with apparent ridges and raised exine regions usually had smaller volumes, suggesting that these two features are possibly indicative of the dehydrated state of Asian *Gnetum* pollen, whereas a psilate surface between microechini indicates its hydrated state. Previous studies pointed out that *Gnetum* pollen grains contract when dried leading to folding and crumpling of their tectum surface, providing a potential way to protect themselves from damage due to changes in their volume (Wodehouse, 1935; Tekleva, 2016). Numerous examples of 'dry *Gnetum* pollen samples' with apparent ridges and/or inflated exine areas have been shown in many studies (Yao *et al.*, 2004; Halbritter, 2015; Tekleva, 2016). In two other gnetalean plants, *Ephedra* and *Welwitschia*, even though their pollen is polyplicate, which already provides a harmomegathic function (Traverse, 2007), ridge–like folds similar to those of *Gnetum* were also found on their plicae (Ickert-Bond, Skvarla & Chissoe, 2003, plate I, fig. 5, plate II, fig. 12, 13; Doores, Osborn & El-Ghazaly, 2007, fig. 8C; Bolinder *et al.*, 2015, figs 4K, 5A, C, F, 6A; Halbritter, 2016). Moreover, two types of ridge, straight and sinuous, were discovered in *Ephedra* pollen (Ickert-Bond, Skvarla & Chissoe, 2003, plate II, figs 12, 13), whereas in *Welwitschia* pollen, a hinge-like structure was observed directly over the furrow (Osborn, 2000, fig. 13). Such folding structures were also recognized in fossil ephedroid pollen (Osborn *et al.*, 1993, plate II, fig. 4–6). Theoretically, pollen exhibit morphological alterations, including exine expansion and contraction, folding of the exine and modification of the lacuna volume to accommodate differences in internal osmotic pressure during wet and dry cycles (Wodehouse, 1935; Blackmore & Barnes, 1986; Hesse *et al.*, 2009). The first two processes are likely to contribute to the harmomegathy of *Gnetum* pollen in our study.

In this present study, the harmomegathic characters of *Gnetum* pollen in their dehydrated and hydrated states were found in both acetolyzed and non-acetolyzed grains. This indicated that the exine elasticity is still effective even in acetolyzed pollen from old herbarium material. Moreover, besides the crumpled appearance exhibited by some dry pollen, the morphology of acetolyzed and non-acetolyzed pollen grains in the same harmomegathic state was generally similar. This is not in agreement with earlier reports that acetolyzed pollen grains of other plant species normally do not show the same shape as those of the hydrated pollen and that the typical infolding present in fresh samples of some pollen species was not found in those from old herbarium specimens (Halbritter & Hesse, 2004). The different appearances between acetolyzed and hydrated pollen from Acanthaceae and Araceae were shown by Scotland *et al.* (1990) (SEM) and Weber, Halbritter & Hesse (1999) (LM).

The harmomegathic mechanism is found in a wide range of fossil gymnosperm pollen from the Upper Palaeozoic and Early Mesozoic eras. The corpus shows strips of exine (taeniae) separated by grooves (striae) allowing the pollen to fold onto itself when drying out during pollination. This feature can still be found in the extant dicot *Nilgirianthus warrensis* (Dalzell) Bremek. (Acanthaceae). In angiosperms, such foldings of pollen exine appear in the membranous colpus area, allowing the pollen to become sealed (Traverse, 2007). Since there is no colpus in the ulcerate (mono-porate) and inaperturate grains, they exhibit different approaches to regulating pollen volume. During dehydration, their pollen walls fold into a mirror-buckled shape (Katifori *et al.*, 2010). This study and previous studies, observed a cup-like feature in dry *Gnetum* pollen (Yao *et al.*, 2004, figs 11, 12; Tekleva, 2016, fig. 1d–f). Therefore, in addition to the raised exine area and the ridge, this mirror-buckling character would be another indicator of dehydration in this pollen group. Halbritter & Hesse (2004) proposed that infolding patterns of pollen were controlled by many factors, such as the aperture condition, the thickness of the pollen wall, thickening and thinning of wall strata, pollen size, ornamentation and pollenkitt. Thus, it would be difficult to envision the shape of the dehydrated pollen grain. For example, even though pollen from *Echinops* L. (Asteraceae) and *Gnetum* show similar exine sculpture (echinate), their harmomegathic characters are different since the folding of *Echinops* pollen occurred in the apertural areas (Garnatje & Martin, 2007, fig. 1A and B).

When water passes through the exine stratum in plant spores and pollen, it does so without the development of any structural change in the exine and it is then absorbed by the intine, stimulating the first state of increased spore and pollen volume. In the second state, the ability of sporomorphs to increase their volume to accommodate water uptake by the intine and cytoplasm is determined by exine elasticity (Pacini, 1990). According to ultrastructural studies, many structures prevent pollen destruction during hydration, e.g. the columellate infratectal layer

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in Boraginaceae, the spongy endexine in *Adansonia* L. or the fragmentation of the foot–layer in many plant groups (Hesse, 2000; Volkova *et al.*, 2013; Rasoamanana *et al.*, 2015). It is possible that the exine flexibility of *Gnetum* pollen found in this present study is facilitated not only by the concentration of infratectal granules between the lamellate endexine and the central part of the raised exine region shown in Hesse (1980), Yao *et al.* (2004) and Tekleva (2016), but also by the loose connection between the individual exine layers. In zona-aperturate pollen grains of *Zamioculcas* Schott and *Gonatopus* Engl. the lamellate endexine accommodates morphological plasticity in response to harmomegathic stress (Hesse *et al.*, 2001). Hesse (2000) mentioned that not only the space within the infratectal stratum, but also the separation of exine units plays a part in pollen volume regulation. However, the ultrastructure of *Gnetum* pollen walls in different states of hydration and dehydration has not yet been examined. Therefore, further studies using TEM are necessary for a better understanding of *Gnetum* palynology and the harmomegathic mechanisms of pollen of the species.

CONCLUSION

The pollen morphology of all the *Gnetum* spp. is generally similar. Microechini on a psilate surface appears to be the character most representative of the hydrated state, whereas the dehydrated state is characterized by raised exine regions and ridge-like folds. Since harmomegathy can cause morphological plasticity, some overlapping of qualitative and quantitative morphological characters occurs between different species and among samples of the same species prepared by different techniques. Therefore, it was not possible to establish any systematic significance in these eight studied pollen species. However, the suggestion that Asian *Gnetum* taxa are monophyletic, which is based on the molecular phylogenetic data, is supported by the existence of the continuous tectum observed in all the *Gnetum* pollen in the present study. Moreover, the authors conclude that comparative palynological study of the variation in the pollen characters of this plant group must take into account the influence of preparation techniques as well as the numbers and conditions of the plants studied.

ACKNOWLEDGEMENTS

The authors are grateful to the Herbarium of Department Biology, Faculty of Science, Prince of Songkla University, the Herbarium of Department of Biology, Faculty of Science, Chang-Mai University and the Queen Sirikit Botanic Garden Herbarium, Botanical Garden Organization, Thailand for providing *Gnetum* specimens, making this project possible. Mr Thomas Duncan Coyne is also thanked for improvements to the English. This project was funded by the Plant Genetic Conservation Project under the Royal Initiative of Her Royal Highness Princess Maha Chakri Sirindhorn, project number SCI590816S.

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APPENDIX

Examined herbarium specimens:

1. *G. cuspidatum* Blume, Sirirugsa, P., Ban Tung Lung, Songkhla, Thailand, Specimen number 184064, PSU herbarium.
2. *G. latifolium* Blume var. *funiculare* (Blume) Markgr., Maxwell JF., Khao Luang National Park, Nakhon Si Thammarat, Specimen number 184038, PSU herbarium; Sirirugsa, P., Doi Phu Kha National Park, Pua, Nan, Specimen number 17664, QBG herbarium.
3. *G. leptostachyum* Blume, Maxwell JF., Chiang Mai, Thailand, Specimen number 01953, CMU herbarium.
4. *G. gnemon* L. var. *tenerum* Markgr., Muangyen N., Chiang Mai, Specimen number 71989, gifted from QBG herbarium; Newman M., Khao Luang National Park, Nakhon Si Thammarat, Specimen number 184034, PSU herbarium; Romkloa Botanical Garden, Specimen number 62505, QBG herbarium.
5. *G. macrostachyum* Hook.f., Maxwell JF., Nakhonnayok, Thailand, Specimen number 20117, CMU herbarium.
6. *G. montanum* Markgr., Maknoi, C., Si satchanalai, Sukhothai, Specimen number 82654, gifted from QBG herbarium, PSU herbarium; La-ongsri W, Tatiya P, Satatha S., Doi Pha Chang, Nan, Specimen number 0015257, gifted from QBG herbarium, PSU herbarium.
7. *G. tenuifolium* Ridl., Sirirugsa, P., Kho Hong hill, Songkhla, Specimen number 184048; 184049, PSU herbarium.

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